Siemens LOGO! PLC — Ladder Logic Specification

📄 **Print-ready PDF:** `docs/pdf/plc-ladder-logic.pdf`

This document specifies all function blocks for the LOGO! Soft Comfort program. Program the LOGO! from this spec before connecting the rig to the RPi HMI.

LOGO! Model

LOGO! 12/24RCE (6ED1052-1MD08-0BA1) - Supply: 12/24V DC - Inputs: 8 digital (24V DC), 4 analog - Outputs: 4 relay (250V AC, 10A each) - Expansion: up to 24 I/O via expansion modules - Ethernet: built-in for LOGO! Web Server + Modbus TCP (port 502) - Software: LOGO! Soft Comfort v8.x (included on CD)

I/O Assignment

Terminal	Type	Signal name	Description	Switch type
I1	Input	ESTOP	E-stop mushroom button	NC
I2	Input	DOOR_INTERLOCK	Chamber door reed switch	NC
I3	Input	OVERTEMP_HW	120°C bimetallic thermostat cutout	NO
I4	Input	MANUAL_START	Green momentary start button	NO
I5	Input	MANUAL_STOP	Red momentary stop button	NC
I6–I8	Input	SPARE	Reserved for future expansion	—
Q1	Output	HEATER_RELAY	SSR control — heater circuit	Relay NO
Q2	Output	SERVO_POWER	24V→5V BEC power enable for servos	Relay NO
Q3	Output	DUT_POWER	DUT power enable relay	Relay NO
Q4	Output	ALARM_BEACON	External 24V alarm beacon	Relay NO

Modbus TCP Register Map

Enable Modbus server in LOGO! Soft Comfort: **Tools** → **Ethernet** → **Modbus server** → **Enable**.

Software enable inputs (written by RPi — M marker coils)

The RPi writes these marker flags as "software requests". The ladder ANDs each one with the hardware safety gate before driving the physical relay. This ensures the E-stop, door interlock, and overtemp cutout always take precedence over software.

Modbus address	Access	LOGO! variable	Description
Coil 2	R/W	M3	SW_HEATER — RPi software heater request → gates Q1
Coil 3	R/W	M4	SW_SERVO — RPi software servo-power request → gates Q2
Coil 4	R/W	M5	SW_DUT — RPi software DUT-power request → gates Q3

Physical output read-back (Q coils — read only from RPi)

Modbus address	Access	LOGO! variable	Description
Coil 8192	R	Q1	Heater relay actual state (read-back)
Coil 8193	R	Q2	Servo power relay actual state
Coil 8194	R	Q3	DUT relay actual state
Coil 8195	R	Q4	Alarm beacon actual state (ladder-only)

Physical input monitoring (discrete inputs — read only)

Modbus address	Access	LOGO! variable	Description
Input 8192	R	I1	E-stop physical state (1=safe)
Input 8193	R	I2	Door interlock (1=closed)
Input 8194	R	I3	HW overtemp (1=tripped)
Input 8195	R	I4	Manual start button (1=pressed)
Input 8196	R	I5	Manual stop button (1=not-pressed=safe)

Function Block Program

Network 1 — Safety gate (internal flag M1)

M1 is HIGH only when ALL physical safety conditions are met. M1 feeds into all output rungs.

```
FBD Network 1:
  [I1 NO contact] ─┬───────────────────────────────────┐
  [I2 NO contact] ─┤                                   ├─►(AND gate) ─►(M1)
  [I3 NC contact] ─┘   (I3 = NC because it's a NO HW switch
                         that makes on fault, so we use NC contact
                         to get "safe = switch open = contact closed logic")
```

Note: I1 and I2 are NC switches → when safe, they provide 24V → LOGO! reads HIGH → use NO contact in ladder. I3 is NO switch → when tripped, provides 24V → use NC contact so M1 goes LOW when I3 is HIGH (tripped).

Network 2 — Run flag SR latch (M2)

```
SET input:   [I4 NO] (manual start button)
RESET input: [I5 NC inverted] OR [M1 inverted] (stop button OR safety gate lost)
Output:      ─►(M2 = "run permitted")
```

Network 3 — Q1 Heater relay

```
[M1 NO] ─[M2 NO] ─[M3 NO (SW_HEATER)]─►(Q1)
```

Heater energises only when: safety OK AND run latched AND RPi software requests it (M3).

Network 4 — Q2 Servo power relay

```
[M1 NO] ─[M4 NO (SW_SERVO)]─►(Q2)
```

Servo power requires only the safety gate and software request (M4). No run latch needed — servos may be positioned during idle or cool-down.

Network 5 — Q3 DUT power relay

```
[M1 NO] ─[M2 NO] ─[M5 NO (SW_DUT)]─►(Q3)
```

DUT power requires all three conditions like the heater. M5 is the software request written by the RPi.

Network 6 — Q4 Alarm beacon

```
[I1 NC inverted] ─OR─[I2 NC inverted] ─OR─[I3 NO]─►(Q4)
```

Alarm fires immediately on any safety fault, independent of all software state. Cannot be silenced by writing M markers — only clearing the hardware fault extinguishes it. - NOT(I1) = alarm when E-stop tripped (I1 LOW → inverted → HIGH → alarm) - NOT(I2) = alarm when door open (I2 LOW → inverted → HIGH → alarm) - I3 = alarm when HW overtemp active (I3 HIGH → alarm)

LOGO! Soft Comfort Setup Procedure

1. Install LOGO! Soft Comfort from the CD included with the PLC.
2. Open Soft Comfort → **File** → **New** → select **LOGO! 8 / 12/24RC**.
3. Set the PC's network adapter to the same subnet as the LOGO! (default: 192.168.0.x).
4. Connect USB programming cable (or Ethernet if your LOGO! model supports it).
5. **Tools** → **Ethernet settings** → **IP address**: set to `192.168.1.100` (or match `rig_config.json`).
6. **Tools** → **Ethernet** → **Modbus server**: enable, port 502.
7. Enter the FBD program as specified in Networks 1–6 above.
8. **File** → **Transfer to LOGO!** — wait for transfer confirmation.
9. Set LOGO! to **RUN mode** via the front-panel button or Soft Comfort.
10. Verify: press Manual Start (I4) → M2 should latch. Check with Soft Comfort online monitor.

Verification Checklist

Before connecting the heater or DUT:

- [ ] LOGO! powered up, showing RUN on display
- [ ] Modbus TCP responding on port 502 (test with `logo_driver.py ping()`)
- [ ] I1 reads HIGH (E-stop button released, NC circuit closed)
- [ ] I2 reads HIGH (door closed)
- [ ] I3 reads LOW (HW thermostat under 120°C, NO contact open)
- [ ] Pressing Manual Start → M2 latches → Q1/Q3 coils can be enabled via software
- [ ] Pressing E-stop → I1 goes LOW → M1 goes LOW → Q1, Q3 de-energise immediately
- [ ] Q4 alarm fires when E-stop pressed
- [ ] Modbus write to Q1 coil with safety gate open → Q1 relay clicks
- [ ] Modbus write to Q1 coil with E-stop pressed → Q1 does NOT energise

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